

CN4119 Final Year Design Project

Cost Estimation – Manufacturing Cost

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Recommended Readings

Turton et al.'s book (Reference 5) Chapter 8

Overview

- ❑ **Cost of manufacture**
 - **Direct manufacturing costs**
 - **Fixed manufacturing costs**
 - **General manufacturing expenses**

- ❑ **Cost of operating labor**

- ❑ **Utility costs**

- ❑ **Raw material costs**

Introduction

- ❑ Costs associated with day-to-day operation of a chemical plant must be estimated
 - B4 economic feasibility of a proposed process can be assessed
- ❑ We need:
 - Process information provided on the PFD
 - An estimate of FCI (that is, C_{TM} or C_{GR})
 - An estimate of number of operators required to operate the plant
- ❑ ***Manufacturing costs => Dollars per unit time***

- ❑ Cost of Manufacture:

COM =

Direct manufacturing costs (DMC)
+ Fixed manufacturing costs (FMC)
+ General expenses (GE)

- ❑ DMC includes many factors

(See the table on the next slide)

- ❑ DMC depends on and varies with ***production rate***

Direct Manufacturing Costs

Factor	Symbol/Equation	Remarks
Raw materials	C_{RM}	Require flow rates & prices
Waste treatment	C_{WT}	Require quantity & treatment cost
Utilities	C_{UT}	Fuel, electricity, steam, water, refrigeration, instrument air, N ₂
Operating labor	C_{OL}	Cost of plant operators
Direct supervisory & clerical labor (DSCL)	(0.1 to 0.25) of C_{OL}	Cost of administrative, engineering & support staff
Maintenance & repairs (M&R)	(0.02 to 0.1) of FCI	Cost of labor & materials
Operating supplies	(0.1 to 0.2) of M&R	Lubricants, chemicals, filters, safety gear & uniforms for operators
Laboratory charges	(0.1 to 0.2) of C_{OL}	Tests for product quality control & troubleshooting
Patents & royalties	(0 to 0.06) of COM	Cost of using licensed technology

❑ Using mid-value of the range:

$$DMC = C_{RM} + C_{WT} + C_{UT} + 1.33 C_{OL} + 0.069 \text{ FCI} + 0.03 \text{ COM}$$

Fixed Manufacturing Costs

- ❑ Not affected by production rate

Factor	Symbol/Equation	Remarks
Depreciation	0.1 FCI	Costs associated with the physical plant, approximate, depends on local tax laws
Taxes and insurance	(0.014 to 0.05) of FCI	Costs associated with property taxes and liability insurance, depends on local taxes
Plant overhead costs	(0.5 to 0.7) of (C_{OL} + DSCL + M&R)	Auxiliary facilities - accounting, fire protection, safety & medical services; canteen & recreation; general engineering

$$FMC = 0.708 C_{OL} + 0.068 FCI + \text{depreciation}$$

General Manufacturing Expenses

- ❑ Costs associated with management level and administrative activities not directly related to the manufacturing process (aka overhead burden)

Factor	Symbol/Equation	Remarks
Administration costs	0.15 of (C_{OL} + DSCL + M&R)	Costs for administration, salaries, buildings, etc.
Distribution & selling costs	(0.02 to 0.2) of COM	Marketing expenses
Research & development	0.05 COM	For enhancing the process and products

$$GE = 0.177 C_{OL} + 0.009 FCI + 0.16 COM$$

Cost of Manufacture

How to Estimate Cost of Manufacture?

- ❑ Add DMC, FMC and GE, and re-arrange to solve for COM:

$$\text{COM} = 0.28 \text{ FCI} + 2.73 C_{OL} + 1.23 (C_{RM} + C_{WT} + C_{UT})$$

- ❑ COM without depreciation:

$$\text{COM}_d = 0.18 \text{ FCI} + 2.73 C_{OL} + 1.23 (C_{RM} + C_{WT} + C_{UT})$$

Do you need to calculate all the components given on Slides 4-6?

Example 8.1: COM_d

Compute COM excluding depreciation for a 92,000 tonne/year nitric acid plant

- ❑ $COM_d = \$14,245,000/\text{year} (= \$155/\text{tonne})$
- ❑ $DMC = 76\% \text{ of } COM_d$
- ❑ $FMC = 7\% \text{ of } COM_d$
- ❑ $GE = 17\% \text{ of } COM_d$
- ❑ Is use of mid-range values for the estimating factors acceptable?

Cost of Operating Labor

Cost of Operating Labor

- ❑ Assuming one operator per shift:
 - 5 eight-hour shifts per week for 49 weeks per year, that is, 245 shifts per operator per year
 - 1095 operating shifts per year required (365 days/year and 3 shifts/day)
 - Number of operators needed to provide this number of shifts is **~4.5 operators**

- ❑ Number of **operators per shift** required to operate the process plant is:

$$N_{OL} = [6.29 + 31.7 P^2 + 0.23 N_{np}]^{0.5}$$

Where,

- P = Number of processing steps involving particulate solids
- N_{np} = Number of non-particulate processing steps, that is, number of equipment handling fluids (compressors, towers, reactors, heaters & exchangers)
- P and N_{np} counted based on the PFD
- P zero for fluid-processing plants, and N_{np} does not include pumps, vessels and tanks

Hence, the total number of operators required per year is $4.5N_{OL}$

Example 8.2: Operating Labor Costs

Compute the operating labor requirement (number of operators required per shift) and annual labor cost for the toluene hydrodealkylation process with compressors, exchangers, furnaces, pumps, reactors, towers and vessels

- ❑ Number of equipment, $N_{np} = 1 \text{ compressor} + 7 \text{ exchangers} + 1 \text{ heater/furnace} + 1 \text{ reactor} + 1 \text{ tower (exclude pumps and vessels)} = 11$
- ❑ Operating labor per annum ≈ 14 operators

Utility and Raw Material Costs

Utility Costs

Typical data taken from Table 8.3 in Turton's book (2013 edition)

Utility		Cost (US \$)
Air supply (3.3 barg)		\$0.35/100 std m^3
Steam – LPS (5 barg, 160°C)		\$27.70/1000 kg
Steam – MPS (10 barg, 184°C)		\$28.31/1000 kg
Steam - HPS (41 barg, 254°C)		\$29.97/1000 kg
Cooling water at 30°C (returned at 40-45°C)		\$14.8/1000 m^3
Process water		\$0.067/1000 kg
Chilled water at 5°C (returned at 15°C)		\$0.185/1000 kg
Potable/drinking water		\$0.26/1000 kg
Boiler feed water (available at 115°C)		\$2.45/1000 kg
Electric power		\$0.06/kWh
Fuel oil		\$549/ m^3
Natural gas		\$0.42/std m^3
Waste water treatment:	Primary (filtration)	\$41/1000 m^3
	Secondary (filtration & activated sludge)	\$43/1000 m^3
	Tertiary (secondary & chemical processing)	\$56/1000 m^3
Solid & liquid waste disposal:	Non-hazardous	\$36/tonne
	Hazardous	\$200-2000/tonne

A Word on the Utility Cost Factors

- ❑ Cost of utilities directly influenced by **cost of fuel**
- ❑ Utilities can be **supplied** in many ways:
 - Purchase from a public/private utility generator
 - Off-site facility providing utilities for various processes at a common location
 - Self-generation
- ❑ The cost factors on the previous slide assume **capital investment** to build a facility for supplying the utility (e.g., cooling tower, steam boiler, etc.) has already been made
 - **Remember grass roots cost (GRC) takes care of this**
- ❑ The cost factors represent the **operating costs to generate the utility**
- ❑ Cost factors given in two units: **\$/GJ** and **\$/common unit (Please refer the text book)**

Utility systems: *Extensive and complex*

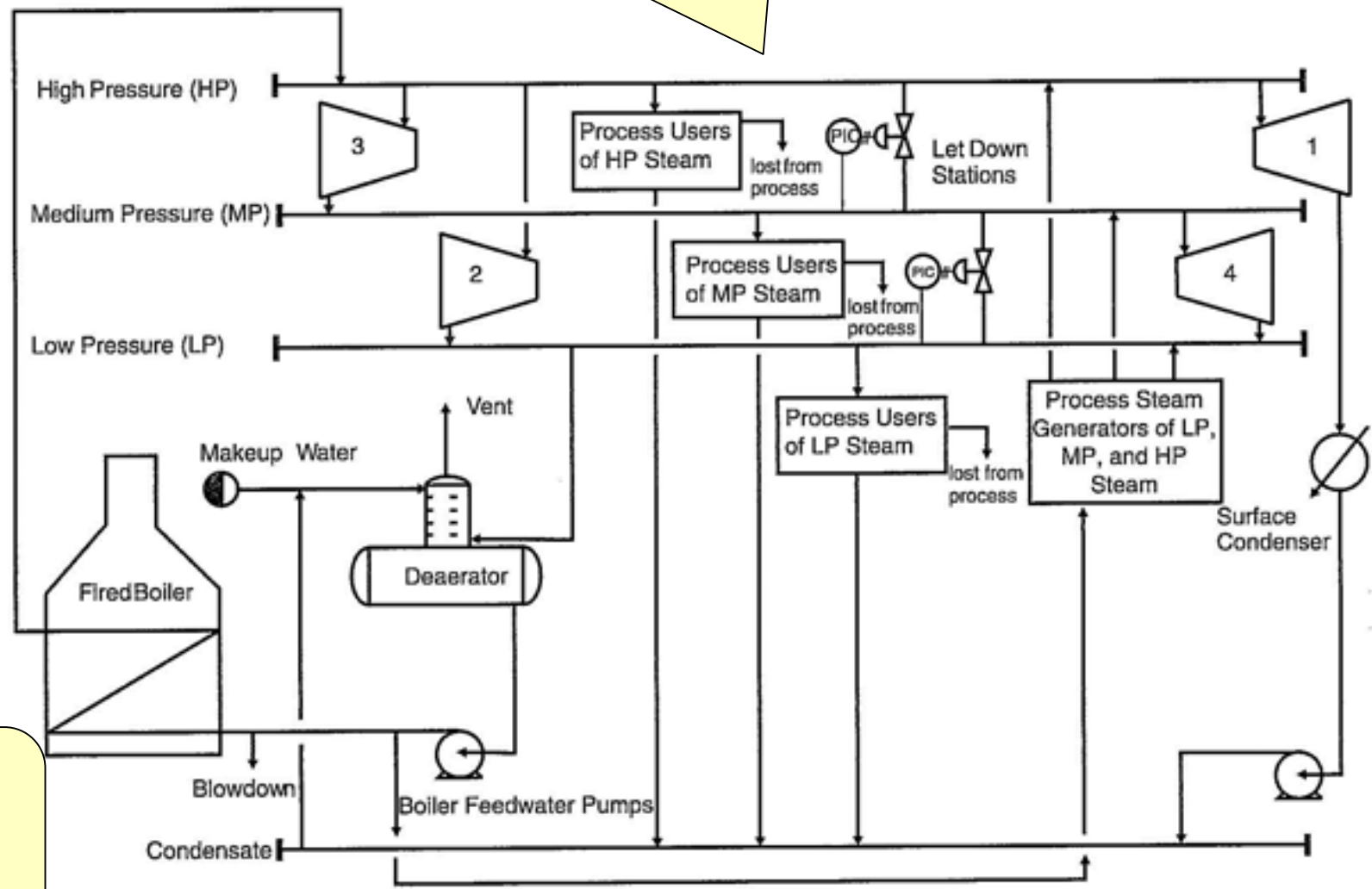


Figure 8.6 Typical Steam Producing System for a Large Chemical Facility

*Do check out
Example 8.6 for
steam costs!*

Allocation of Utility

- ❑ Allocation of appropriate utility is very important
 - The driving force for heat transfer must always be positive
- ❑ Recall the **heat transfer equation**
$$Q = U A LMTD$$
 - Q is from HYSYS simulation
 - U is from literature for the respective services
 - $LMTD$ depends on allocation of utility (unless it is process-to-process heat transfer)
 - A can then be calculated

A is needed for capital cost, Q is needed for operating cost

- ❑ Let us consider two scenarios with minimum temperature driving force of 10°C
 - Heating a stream from 100°C to 155°C

 - Cooling a stream from 100°C to 35°C

Raw Material Costs

- ❑ Cost of raw materials can be estimated by using the current price listed in publications such as *Chemical Market Reporter*
- ❑ See Table 8.4 for selling prices of some common chemicals (as of August 2008)
- ❑ Largest operating cost usually raw materials => **Accurate price data important**

Table 8.4 Costs of Some Common Chemicals*

Chemical	Cost (\$/kg)	Typical Shipping Capacity or Basis for Price
Acetaldehyde	1.003	Railroad Tank Cars
Acetic Acid	0.838	Railroad Tank Cars
Acetone (MMA grade)	1.102	Railroad Tank Cars
Acrylic Acid	1.918	Railroad Tank Cars
Allyl Chloride	1.80 [†]	F.O.B. Gulf Coast
Benzene	0.919	Barge, Gulf Coast
Chlorine	0.284	Railroad Tank Car
Dimethyl Ether	0.841 [†] (2011)	Railroad Tank Car
Ethanol (190 Proof)	1.138	Railroad Tank Car
Ethylbenzene	1.268	Railroad Tank Car, Gulf Coast
Ethylene	1.488	Contract
Ethylene Oxide	1.687	Railroad Tank Car
Formaldehyde/Formalin (37 wt%)		
No-inhibitor	0.838	Railroad Tank Car, Gulf Coast
7% Methanol Inhibitor	0.463	Railroad Tank Car, Gulf Coast
Hydrochloric Acid (23%Be)	0.090	Railroad Tank Car, Works
Iso-Butylene	0.706	F.O.B. Works
Iso-Propanol (99%)	1.400	Railroad Tank Car
Maleic Anhydride	1.158	Railroad Tank Car
Methanol	0.672	F.O.B. Gulf Coast
Methyl Ethyl Ketone	1.598	Railroad Tank Car
MTBE	0.609	Barge, F.O.B. Asia
Propylene (Polymer Grade)	1.488	F.O.B. Gulf Coast—Spot Price

Table 8.4: Costs of some common chemicals
(Please check latest version for updated figure)

Yearly Costs

- ❑ Manufacturing and associated costs in terms of \$/year
- ❑ Information on PFD in terms of kg or kmol per hour
- ❑ To compute yearly costs of raw materials or utilities, fraction of time the plant is operating in a year must be known
 - This is called ***stream factor (SF)***
 - $$SF = \frac{\text{Number of days of operation}}{365}$$
 - Typical values of SF = 0.90 to 0.96
- ❑ SF is only for C_{RM} , C_{UT} and C_{WT}

For design project, you will be given number of hours of operation per annum

Example 8.9: Estimating Utility Costs

Estimate quantities and yearly costs of utilities for several equipment in a toluene hydrodealkylation plant assuming stream factor = 0.95

- ❑ Heaters/coolers based on duty (GJ/hr) and price of HPS/MPS/LPS/cooling water per GJ

- ❑ Furnace based on duty (GJ/hr) and price of natural gas per GJ (based on lower heating value of natural gas)

- ❑ Compressor (using electric drive)/pumps based on power (kW) and price of electricity per kW

Self-study Example

Example 8.10: COM_d for a Complete Plant

Estimate cost of manufacture without depreciation (COM_d) of benzene from toluene

❑ Raw materials: Toluene = \$86 million/year

Hydrogen = \$6.6 million/year

Total = \$92.6 million/year

Raw materials make up the bulk of the operating cost

❑ Utilities: Steam = \$3.4 million/year

Cooling Water = \$0.2 million/year

Fuel Gas = \$2.8 million/year

Electricity = \$0.04 million/year

Total = \$6.4 million/year

❑ No waste streams and waste treatment cost

❑ Operating labor: \$0.8 million/year

❑ Fixed capital investment, FCI = \$12.5 million

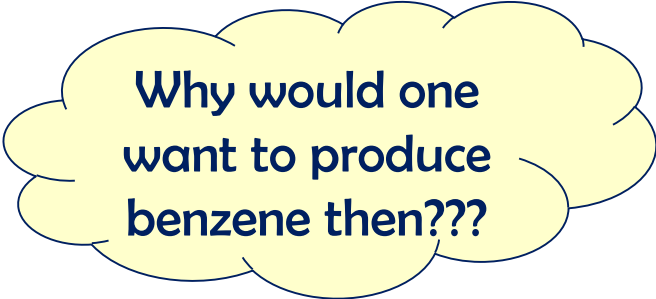
$$\begin{aligned}\text{❑ } \text{COM}_d &= 0.18 \text{ FCI} + 2.73 C_{OL} + 1.23 (C_{RM} + C_{WT} + C_{UT}) \\ &= \$126.3 \text{ million/year}\end{aligned}$$

❑ Benzene production = 8,210 kg/hr

$$= 8,210 \times 24 \times 365 \times \mathbf{0.95} = 68.3 \text{ million kg/year}$$

❑ Cost of benzene production = \$1.85/kg

⇔ Compare with benzene price (\$0.919/kg from Table 8.4)



Why would one
want to produce
benzene then???

Manufacturing Cost: Summary

- ❑ Cost of manufacture depends on:
 - Fixed capital investment
 - Cost of operating labor
 - Cost of utilities
 - Cost of waste treatment
 - *Cost of raw materials* - usually the biggest cost
- ❑ Amount of raw materials and utilities can be directly obtained from the PFD
- ❑ Cost of operating labor can be estimated from the number of pieces of equipment given on the PFD